

Artificial Intelligence

Machine Learning Fundamentals

Lecture 02 and Machine

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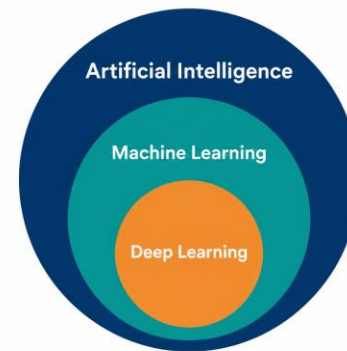


Lecture overview



This lecture develops machine learning fundamentals from its basic definition to its mathematical operation and practical use. The explanation follows a gradual sequence so that each new idea connects with the previous one.

The main discussion covers Machine Learning, Programming and Machine Learning, AI, Machine Learning, and Deep Learning, Machine Learning pipeline. Numerical or practical examples show how the concepts produce observable outputs.



Example of the learning sequence: identify the inputs, apply the stated rule or equation, calculate the output, and interpret the result within the limits of the model.



The algorithm receives examples and learns a model. The model maps new inputs to predictions. The programmer defines the learning process rather than every decision rule. Training data produces Model; Model and New data produce Prediction. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

A small numerical test should use known inputs and an expected output. Comparing the computed result with the expected result reveals whether the method and its assumptions were applied correctly.



Traditional programming starts with rules. Machine Learning starts with examples and desired outcomes. Both still require software, testing, and careful problem definition. Spam detection: learned patterns adapt better than one fixed keyword rule. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

A fixed rule can label a student as Pass when grade ≥ 50 . Machine learning is more suitable when success depends on several interacting features whose relationship is not known in advance.

AI, Machine Learning, and Deep Learning



Artificial Intelligence is the broad field. Machine Learning is a family of data-driven AI methods. Deep Learning uses neural networks with multiple layers. AI contains ML; ML contains Deep Learning. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

Early CNN layers may detect edges, middle layers combine edges into shapes, and deeper layers respond to object parts such as eyes or wheels.

Machine Learning pipeline and Core terminology



Collect representative data. Train a model. Prepare and clean the data. Test with unseen data. Choose inputs called features. Use the model for prediction. Feature: an input variable. Label or target: the value to predict. Model: the learned mathematical mapping. Training: estimating model parameters. Prediction: output for a new input. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

A small numerical test should use known inputs and an expected output. Comparing the computed result with the expected result reveals whether the method and its assumptions were applied correctly.

Training, validation, and testing



Training data updates model parameters. Validation data supports model selection and tuning. Test data estimates performance on unseen examples. Never tune the model using the final test set. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

With 1,000 records, a simple split may use 700 for training, 150 for validation, and 150 for final testing. The test records remain untouched until model selection ends.

Types of Machine Learning



Supervised learning uses labeled examples. Unsupervised learning searches for structure without labels. Reinforcement learning uses rewards and penalties during interaction. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

A supervised dataset contains images paired with Cat or Dog labels. The model compares its prediction with the correct label and updates its parameters to reduce the error.

Supervised Learning and Classification and Regression



Each training example includes an input and correct answer. The model learns to predict a class or a numerical value. Examples: spam detection, diagnosis support, and price estimation. Classification predicts a category. Regression predicts a continuous number. The required output determines the problem type. Category: Pass or Fail Number: House price. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

Email spam detection is classification because the output is a category. Predicting a house price is regression because the output is a continuous number.

Unsupervised Learning



Training examples have no predefined labels. Clustering groups similar observations. Dimensionality reduction creates a compact representation. Customer segmentation is a common clustering example. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

A supervised dataset contains images paired with Cat or Dog labels. The model compares its prediction with the correct label and updates its parameters to reduce the error.



An agent selects an action in an environment. The environment returns a state and reward. The agent learns a policy that increases long-term reward. Correct move: positive reward Collision: penalty. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

A robot receives a positive reward for reaching a target and a negative reward for collision. Repeated interaction increases the probability of actions that produce higher total reward.

Student performance and Evaluation metrics



Features: study hours, attendance, previous grade. Target: Pass or Fail. Learning type: supervised learning. Task: classification. New student: 5 hours, 85% attendance, grade 70; model predicts Pass. Accuracy measures the fraction of correct classifications. Precision asks how many predicted positives were correct. Recall asks how many real positives were found. Mean Absolute Error measures average numerical error. Choose a metric that matches the cost of mistakes. Evaluation must use data that did not guide parameter updates. The selected metric should reflect the real cost of false positives, false negatives, and numerical prediction errors.

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Benefits and risks of Machine Learning



Benefits: fast analysis, repeatable predictions, and discovery of patterns in large datasets. Benefits depend on representative data and a clearly defined task. Risks: biased samples can create unfair predictions. Risks: leakage, incorrect labels, and weak testing can produce misleading accuracy. A useful model must combine predictive performance with reliable evaluation. Learning changes model parameters so that future outputs better match the required behavior. Reliable training requires representative data, a measurable error, and a stopping condition that prevents unnecessary fitting.

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Classroom activity



House price prediction: identify the learning type. Spam detection: identify the learning type. Customer grouping: identify the learning type. Robot navigation: identify the learning type. Answers: Regression; Classification; Clustering; Reinforcement Learning. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

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Bridge to neural networks



Neural networks are Machine Learning models inspired by biological neurons. They combine weighted inputs and activation functions. Later lectures develop Hebb, Perceptron, Adaline, Madaline, Backpropagation, and CNN. The concept should be understood as a sequence of inputs, mathematical processing, and outputs. Clear assumptions and suitable data are necessary before the result can be interpreted correctly.

A strong biological synapse has a larger influence on the next neuron. In the artificial model, a larger weight gives the corresponding input greater influence on the net value.



A complete application of machine learning fundamentals begins with a clearly defined input and target. The method applies its architecture, learning rule, or processing equation to obtain a measurable output. Evaluation compares this output with the expected response, quantifies the error, and checks whether the model generalizes beyond the training examples. The interpretation should also consider the assumptions of the algorithm and the quality of the input data.

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Lecture summary



The lecture connected machine learning, programming and machine learning, ai, machine learning, and deep learning with machine learning pipeline, core terminology, training, validation, and testing. These concepts describe how an intelligent or neural system represents inputs, performs a calculation, learns from error or association, and produces a result.

The central principle is that a model should be understood through its inputs, parameters, mathematical operation, output, and evaluation method. Each part affects the reliability of the final result.

Example connection: a correct calculation is useful only when the data and assumptions represent the real task.